MIL-T-52418B(ME)
23 March 1978
SUPERSEDING
MIL-T-52418A(ME)
27 July 1971

MILITARY SPECIFICATION

TENSIOMETER: GAGE INDICATING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a gage-indicating tensiometer.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.

FSC 6635

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

D5293-1	- Tensiometer, Gage Indicating.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The tensiometer shall be as shown on Drawing D5293-1 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawing to insure correct fit, assembly, and operation of the tensiometer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

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3.2 First article (first produced tensiometer). The contractor shall furnish one or more tensiometers as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce tensiometers that comply with the requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Performance. The tensiometer gage shall have an accuracy of plus or minus 5 percent when tested as specified in 4.5.2.1.

3.5 Identification marking. The tensiometer shall be identified in accordance with MIL-STD-130.

3.6 Treatment and painting. The portions of the tensiometer normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

3.7 Workmanship.

3.7.1 Castings and forgings. All parts, components, and assemblies of the tensiometer which include castings and forgings shall be cleaned of harmful extraneous material such as sand, dirt, pits, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.7.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

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3.7.3 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the tensiometer to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 3/16 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the tensiometer shall be at the option of the contractor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawing, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced tensiometer shall be cause for disassembly, in the presence of a Government representative, of the first produced tensiometer to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawing in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced tensiometer. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced tensiometer inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced tensiometer inspection.

4.3.1 Examination. The first produced tensiometer shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Test. The first produced tensiometer shall be tested as specified in 4.5.2.1. Failure of the performance test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Samples. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective.

4.4.3 Test.

4.4.3.1 Samples. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The tensiometer shall be examined as specified herein for the following major defects:

101. Dimensions not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Identification marking missing or illegible.
105. Workmanship not as specified.

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4.5.2 Test.

4.5.2.1 Performance. The tensiometer shall be subjected to 1,000 and 2,000 pound loads. Inability of the tensiometer to indicate the loads within an accuracy of plus or minus 5 percent or any evidence of cracks, rupture, or permanent deformation shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106.	Materials, method, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
107.	Interior surfaces of the tensiometer not coated with preservative as specified.
108.	Box closure not Method V as specified.
109.	Strapping not as specified.
110.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. The interior surface of the tensiometer shall be coated with Type P7 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116. The tensiometer shall then be preserved in accordance

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with MIL-P-116, Method III, using a close-fitting box conforming to PPP-B-636, V3c, style optional. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. Tensiometers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Tensiometers preserved as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional, or PPP-B-621, Class 2, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, Finish A, size as applicable.

5.2.2 Level B. Tensiometers shall be packed as specified in 5.2.1 for Level A, except that boxes shall be Domestic Type or class.

5.2.3 Commercial. Tensiometers preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The tensiometer is intended to be used in setting the correct tension in the track cable of the aerial tramway and cableway.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of first produced tensiometer, and number of tensiometers required (see 3.2).
(c) Welding process required if different from contractors proposed process (see 3.7.3).

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(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced tensiometer. Any changes or deviations of production tensiometers from the approved first produced model during production will be subject to the approval of the contracting officer. Approval of the first produced model will not relieve the contractor of his obligation to furnish tensiometers conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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